

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10: 1.4 ENERGY, WORK, POWER AND MACHINES

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Final Explanation: How do pulleys reduce the effort needed to lift water, and how does energy transfer in the process? Imagine you are explaining to a friend in Kitui County why Mwende, who is 10 years old, can lift the bucket with the pulley system while her mother, Mama Kavata Mutua, finds the direct lift exhausting. Use what you observed and learned in Lessons 1 to 8. Use $g = 10 \text{ N/kg}$ everywhere. Show every calculation step, with units, and write full sentences for your explanations. Use the data given in each section. Write your answers in the space under each prompt.

Part 1: What the bucket gains

A 20 kg bucket of water (weight 200 N) is lifted 8 m up a well at a steady speed in 20 s. The water surface is the 0 m level and the top of the well is 8 m above it.

Bucket PE table ($m = 20 \text{ kg}$):

Height above water (m)	PE (J)
0	0
2	400
4	800
6	1200
8	1600

(a) Use $PE = mgh$ to show how the table value for 8 m is found.

- (b) Mama Kavata lifts the bucket by hand. Mwende lifts it with the pulley system. Compare the energy the bucket gains in the two cases and explain why.
- (c) The bucket moves at a steady speed. What happens to its kinetic energy, and where does all of the energy gain go?

Write your answer here:

Part 2: Work, force and distance

Use $W = Fd$, with F in the direction of motion.

- (a) Calculate the work Mama Kavata does lifting the bucket directly by hand through 8 m, and her power if the lift takes 20 s.
- (b) With the pulley system, Mwende pulls the rope with an effort of 125 N and the rope is pulled 16 m while the bucket rises 8 m in 20 s. Calculate the work she does and her input power.
- (c) Compare the force and the distance in the hand lift and the pulley lift. What does the pulley system change, and what does it not change about the work needed?

Write your answer here:

Part 3: Where the energy comes from and where it goes

Use conservation of energy.

(a) Where does the energy come from to lift the bucket, and can the pulley create energy? Explain.

(b) In the pulley lift, the bucket gains 1600 J of PE but Mwende does 2000 J of work. Calculate the difference and explain where this energy goes. Is it destroyed?

(c) In Lesson 4, a real bucket fell from 8 m and hit the water at 11 m/s. Calculate its KE at the water using $KE = \frac{1}{2}mv^2$, and the energy transferred to the surroundings, given that its PE at the top was 1600 J. How does this help you explain your answer to (b)?

Write your answer here:

Part 4: Machines, MA, VR and efficiency

The pulley system has one fixed pulley and one movable pulley, with 2 rope sections supporting the bucket. The load is 200 N and the actual effort is 125 N. The rope is pulled 16 m while the bucket rises 8 m.

- (a) Calculate the MA, the VR and the efficiency of the pulley system.
- (b) Calculate the effort needed if the pulley system had no friction (an ideal pulley system), using the VR.
- (c) A crowbar lifts a 400 N load with an effort of 125 N. The effort arm is 1.2 m and the load arm is 0.3 m. Calculate its MA, VR and efficiency, and say which is more efficient, the crowbar or the ramp that has an efficiency of 62.5%.
- (d) A single fixed pulley has $VR = 1$. What does it do, and why can a child still use it at the well?

Write your answer here:

Part 5: Final explanation

Write a complete explanation (6 to 8 sentences) that answers the driving question: How do pulleys reduce the effort needed to lift water, and how does energy transfer in the process? Include: (i) the energy the bucket gains, (ii) the effort and the rope distance in the hand lift and in the pulley lift, (iii) where the energy comes from, (iv) what happens to the energy that does not become PE, and (v) the statement about conservation of energy. Use the numbers from the Kitui well.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of data and calculations	All calculations (PE, work, power, KE, MA, VR, efficiency) are correct, use $g = 10 \text{ N/kg}$, show steps and give correct units.	Most calculations are correct with steps; one or two small slips in units or arithmetic.	Several calculations are missing, incorrect or have no units or working.
Explaining why the pulley feels easier	Clearly states a smaller force (125 N vs 200 N) over a longer distance (16 m vs 8 m) and that the work needed is not reduced.	States the smaller force and longer distance but is unclear about the work.	Says the pulley makes lifting easier without describing force and distance, or claims the pulley reduces work or gives energy.
Energy source, transfer and conservation	Identifies the chemical energy from food as the source, accounts for $2000 \text{ J} = 1600 \text{ J PE} + 400 \text{ J heat}$, and states that energy is conserved and not destroyed.	Identifies the energy source and the 400 J as heat, but conservation is incompletely stated.	Does not identify the source, or says the energy is lost or destroyed.
Machine terms and efficiency	Correctly defines and calculates MA, VR and efficiency, and compares machines and the fixed pulley accurately.	Calculates most of MA, VR and efficiency correctly with minor errors in comparison.	Confuses MA, VR and efficiency or cannot calculate them.
Scientific communication	The explanation is organised, uses scientific terms accurately and refers to the Kitui well data throughout.	The explanation is mostly clear with some terms used loosely.	The explanation is disorganised or uses few scientific terms.

ATTRIBUTION AND LICENSING

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Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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